

Cloud Computing

Mock Exam 5 | Course ID: noc26_cs55

Q1. What does the ACM definition of Computing encompass?

- A) Only designing hardware and software.
- B) Any goal-oriented activity requiring, benefiting from, or creating computers, including designing hardware/software, processing information, and creating intelligent systems.
- C) Primarily processing information and creating intelligent systems.
- D) Only activities related to high-end scientific research.

Source: Lecture_W1L19_Lecture_1__Cloud_Computing__Overview | Difficulty: Easy

Q2. According to the NIST definition, what is a key characteristic of Cloud Computing regarding resource provisioning?

- A) Resources are provisioned only with significant human interaction.
- B) Resources require long lead times for provisioning.
- C) Resources can be rapidly provisioned and released with minimal management effort or service provider interaction.
- D) Resources are static and cannot be easily changed once provisioned.

Source: Lecture_W1L20_Lecture_2__Cloud_Computing__Overview__contd__ | Difficulty: Easy

Q3. Which cloud service model is analogous to using a ride-sharing service, where you just use it to get to your destination?

- A) Infrastructure as a Service (IaaS)
- B) Platform as a Service (PaaS)
- C) Software as a Service (SaaS)
- D) Anything as a Service (XaaS)

Source: Lecture_W1L21_Lecture_3__Cloud_Computing__Introduction | Difficulty: Easy

Q4. What is the primary method for achieving elasticity in the cloud by adding more machines (nodes) to a resource pool?

- A) Vertical Scaling (Scale-Up)
- B) Horizontal Scaling (Scale-Out)
- C) Resource Pooling
- D) Virtualization

Source: Lecture_W1L22_Lecture_4__Cloud_Computing_Architecture | Difficulty: Easy

Q5. What is the core technology that differentiates cloud computing from the traditional client-server model?

- A) Dedicated servers
- B) Virtualization
- C) Fixed IP addresses
- D) Request-response model

Source: Lecture_W1L23_Lecture_5__Cloud_Computing_Architecture__contd__ | Difficulty: Easy

Q6. Which cloud deployment model is provisioned for open use by the general public and operated by a third-party provider (e.g., AWS, Azure, Google Cloud)?

- A) Private Cloud
- B) Community Cloud
- C) Public Cloud
- D) Hybrid Cloud

Q7. What is the software layer that creates and runs virtual machines (VMs) and sits between the hardware and the virtual machines, managing resource allocation?

- A) Guest OS
- B) Host OS
- C) Hypervisor (VMM)
- D) Cloud Manager

Source: Lecture_W2L29_Lecture_7__Cloud_Computing__Virtualization | Difficulty: Easy

Q8. What is the primary purpose of XML (Extensible Markup Language)?

- A) To display data for human users in web browsers.
- B) To structure and transport data for machines in a text-based, self-describing format.
- C) To define programming logic for web applications.
- D) To manage network protocols for data transmission.

Source: Lecture_W2L30_Lecture_8__Cloud_Computing__XML_Basics | Difficulty: Easy

Q9. What is a formal, legally binding contract between a service provider and a service consumer that defines the level of service, responsibilities, and remedies for non-compliance?

- A) Key Performance Indicator (KPI)
- B) Service Level Objective (SLO)
- C) Service Level Agreement (SLA)
- D) Quality of Service (QoS)

Source: Lecture_W3L37_Lecture_11__Service_Level_Agreement | Difficulty: Easy

Q10. Which OpenStack component is the official web-based user interface (dashboard) for managing services and provisioning resources?

- A) Nova
- B) Neutron
- C) Horizon
- D) Keystone

Source: Lecture_W3L41_Lecture_15__Open_Stack | Difficulty: Easy

Q11. What is the significance of Service Level Agreements (SLAs) in the cloud business model?

- A) They primarily define the marketing strategy for cloud services.
- B) They translate technical capabilities into measurable, contractual obligations, allowing consumers to choose services based on specific needs.
- C) They are only used for internal resource management within the cloud provider's infrastructure.
- D) They dictate the programming languages used for cloud applications.

Source: Lecture_W1L19_Lecture_1__Cloud_Computing__Overview | Difficulty: Medium

Q12. Which computing paradigm introduced the concept of tapping into a vast pool of computing power like a utility, often harnessing underutilized compute cycles, and treated geographically dispersed resources as a single, virtual supercomputer?

- A) Distributed Systems
- B) Cluster Computing
- C) Utility Computing
- D) Grid Computing

Source: Lecture_W1L20_Lecture_2__Cloud_Computing__Overview__contd__ | Difficulty: Medium

Q13. In the IaaS cloud service model, what is the user primarily responsible for managing?

- A) Physical hardware, networking, and storage.
- B) The hypervisor and underlying operating system.
- C) The operating system, applications, and data.
- D) The complete development and deployment environment including databases and web servers.

Source: Lecture_W1L21_Lecture_3__Cloud_Computing__Introduction | Difficulty: Medium

Q14. Which architectural concern, described as a 'vertical stack', is not confined to a single layer (like IaaS or PaaS) but cuts across the entire service stack, from hardware to the application?

- A) Elasticity
- B) Interoperability
- C) Management and Security
- D) Performance

Source: Lecture_W1L22_Lecture_4__Cloud_Computing_Architecture | Difficulty: Medium

Q15. When is PaaS (Platform as a Service) typically ideal to use?

- A) For applications requiring real-time data or extremely fast processing.
- B) For projects where developers want to focus on application logic, not infrastructure management.
- C) For mission-critical operations requiring the absolute highest level of performance and custom hardware.
- D) For situations where regulations prohibit hosting data externally.

Source: Lecture_W1L23_Lecture_5__Cloud_Computing_Architecture__contd__ | Difficulty: Medium

Q16. What is a key characteristic of a Community Cloud?

- A) It is exclusively used by a single organization.
- B) It is open for use by the general public.
- C) It is shared by several organizations with common concerns like security, compliance, or mission.
- D) It is a mix of two or more distinct cloud infrastructures.

Source: Lecture_W2L28_Lecture_6__Cloud_Computing_Architecture__Deploym | Difficulty: Medium

Q17. Which XML parser is best suited for parsing very large XML files or streaming data in real-time due to its low memory footprint and event-based nature?

- A) DOM Parser
- B) SAX Parser
- C) XSLT Processor
- D) XPath Evaluator

Source: Lecture_W2L31_Lecture_9__Cloud_Computing_XML_Basics_II | Difficulty: Medium

Q18. What does the Coefficient of Variation (CV) measure in cloud economics, and what does a lower CV signify?

- A) It measures the absolute peak demand; a lower CV means higher peak demand.
- B) It measures the total cost of ownership; a lower CV means lower costs.
- C) It measures the relative variability of a data series; a lower CV signifies a 'smoother,' more predictable demand.
- D) It measures the network latency; a lower CV means higher latency.

Source: Lecture_W3L38_Lecture_12__Cloud_Economics | Difficulty: Medium

Q19. What is the primary difference in data organization between Bigtable and traditional relational databases?

- A) Bigtable is row-oriented, while relational databases are column-oriented.
- B) Bigtable is designed for small datasets, while relational databases handle massive data.
- C) Bigtable is column-oriented and optimized for analytics and sparse data, while relational databases are typically row-oriented.
- D) Bigtable uses SQL for queries, while relational databases use NoSQL.

Q20. What is the primary goal of MapReduce regarding parallel efficiency and scalability?

- A) To minimize the number of processors used regardless of data size.
- B) To ensure that efficiency decreases as the workload and resources increase proportionally.
- C) To maintain high efficiency as the workload (data size) and resources (number of processors) increase proportionally.
- D) To eliminate the need for fault tolerance in distributed systems.

Source: Lecture_W3L40_Lecture_14__Introduction_to_Map_Reduce | Difficulty: Medium

Q21. Which OpenStack component is responsible for managing and storing virtual machine disk images (e.g., Ubuntu, CentOS) that Nova uses to launch new VMs?

- A) Nova
- B) Neutron
- C) Keystone
- D) Glance

Source: Lecture_W3L41_Lecture_15__Open_Stack | Difficulty: Medium

Q22. In Microsoft Azure's architectural blueprint, what is the primary function of a Worker Role?

- A) To handle incoming user requests and present information to the user.
- B) To manage the underlying physical hardware and virtualization.
- C) To run background processing, distributed tasks, or other complex, non-interactive computations.
- D) To store persistent data in blobs or tables.

Source: Lecture_W4L47_Lecture_17__Cloud_Computing__Case_Study_with_a_Co | Difficulty: Medium

Q23. How do the core principles of Distributed Computing (fault tolerance, resource sharing, scalability) fundamentally enable the cloud computing paradigm?

- A) They allow cloud providers to operate with a single point of failure, simplifying management.
- B) They ensure that cloud resources are always physically located in a single data center for optimal performance.
- C) They provide the fundamental building blocks for high availability, elasticity, and efficiency, allowing the cloud to withstand failures, share resources, and expand on-demand.
- D) They prioritize individual node autonomy over collaboration, making cloud services less interdependent.

Source: Lecture_W1L19_Lecture_1__Cloud_Computing__Overview | Difficulty: Hard

Q24. What are the primary trade-offs or concerns that must be weighed against the advantages (cost, scalability, reliability) of cloud computing?

- A) Reduced need for internet connection and simplified resource management.
- B) Enhanced data privacy and guaranteed performance consistency.
- C) Security, data privacy, performance variability, and dependency on a constant internet connection.
- D) Lower capital expenditure and increased control over infrastructure.

Source: Lecture_W1L21_Lecture_3__Cloud_Computing__Introduction | Difficulty: Hard

Q25. According to the Popek & Goldberg Requirements, what are the three formal conditions for a computer architecture to efficiently support virtualization?

- A) High availability, low latency, and strong encryption.
- B) Equivalence, Resource Control, and Efficiency.
- C) Multi-tenancy, elasticity, and on-demand self-service.
- D) Hardware-assisted, paravirtualized, and full virtualization support.

Source: Lecture_W2L29_Lecture_7__Cloud_Computing__Virtualization | Difficulty: Hard

Q26. Describe the XML processing workflow, specifically differentiating between a 'well-formed' and a 'valid' XML document.

- A) A 'well-formed' document conforms to a schema, while a 'valid' document has correct syntax. The parser first checks validity, then well-formedness.
- B) A 'well-formed' document is syntactically correct (e.g., closed tags, one root), and a 'valid' document is a well-formed document that also conforms to a separate schema (DTD/XSD). The parser checks well-formedness first.
- C) A 'well-formed' document is one that can be displayed in a web browser, and a 'valid' document is one that contains only text. No parsing is needed.
- D) There is no difference; 'well-formed' and 'valid' are interchangeable terms for XML documents.

Source: Lecture_W2L30_Lecture_8__Cloud_Computing__XML_Basics | Difficulty: Hard

Q27. In cloud economics, under what condition is the cloud generally more economical than owning on-premise infrastructure, considering peak demand (P), average demand (A), and utility premium (U)?

- A) When $P / A < U$
- B) When $P = A$
- C) When $P / A > U$
- D) When U is very low, regardless of P/A

Source: Lecture_W3L38_Lecture_12__Cloud_Economics | Difficulty: Hard

Q28. Explain the key difference in parallel efficiency between 'dividing the words' and 'dividing the documents' approaches for a word-counting problem, and why the latter aligns with MapReduce.

- A) 'Dividing the words' is more efficient because each processor scans fewer documents. MapReduce aligns with this by assigning specific words to processors.
- B) 'Dividing the words' is inefficient because processors waste time reading irrelevant documents, leading to decreasing efficiency with more processors. 'Dividing the documents' is efficient because every read is useful, aligning with MapReduce's strategy for scalability.
- C) Both approaches have similar parallel efficiency. MapReduce can use either, depending on the data size.
- D) 'Dividing the documents' is less efficient due to increased communication overhead. MapReduce avoids this by centralizing all data processing.

Source: Lecture_W3L40_Lecture_14__Introduction_to_Map_Reduce | Difficulty: Hard

Q29. Describe the coordinated process of launching a new Virtual Machine (VM) in OpenStack, highlighting the roles of at least three different core components.

- A) Horizon directly launches the VM, then Keystone assigns an IP address, and Glance provides storage.
- B) The user requests a VM via Horizon, Keystone authenticates, Nova schedules it, Glance provides the OS image, and Neutron allocates network resources.
- C) Neutron creates the VM, Cinder provides the OS image, and Nova handles user authentication.
- D) Glance authenticates the user, Nova manages network resources, and Horizon schedules the VM on a physical host.

Source: Lecture_W3L41_Lecture_15__Open_Stack | Difficulty: Hard

Q30. A cloud service guarantees 99.95% availability. Over a 30-day month, with service expected 12 hours/day, the total observed downtime was 9 hours and 40 minutes. If the monthly cost is \$1500, and a 25% service credit is applied if availability is <99%, and 10% if <99.95% but ≥99%, what is the effective cost? (Use the lecture's calculation for 9h 40m downtime as 9.417 hours for consistency).

- A) \$1500 (no violation)
- B) \$1350 (10% credit)
- C) \$1125 (25% credit)
- D) \$750 (50% credit)

Answer Key

1: B	2: C	3: C	4: B	5: B
6: C	7: C	8: B	9: C	10: C
11: B	12: D	13: C	14: C	15: B
16: C	17: B	18: C	19: C	20: C
21: D	22: C	23: C	24: C	25: B
26: B	27: C	28: B	29: B	30: C